

Back Titration Questions:

1. An indigestion tablet was crushed up and an excess of hydrochloric acid (25.0 mL of 1.00 M) added. The resulting solution was titrated against 0.500 M NaOH requiring 25.80 mL of the NaOH. Calculate the mass of calcium carbonate in the tablet.
2. A fertilizer contains ammonium sulphate and potassium sulphate. A sample of 1.455 g of the fertilizer was warmed with 25.0 mL of a 0.200 M solution of sodium hydroxide giving off ammonia gas. The remaining NaOH that was not used required 28.70 mL of a 0.100 M solution of hydrochloric acid for neutralization. Calculate the percentage by mass of ammonium sulphate in the sample.
3. An impure sample of barium hydroxide of mass 1.6524 g was allowed to react with 100 mL of 0.200 M hydrochloric acid. When the excess acid was titrated against 0.228 M sodium hydroxide in a back titration, 10.90 mL of sodium hydroxide solution was required. Calculate the percentage purity of the sample of barium hydroxide.
4. 1.50 g of impure magnesium carbonate is added to 100 mL of 0.400 M hydrochloric acid. The left-over acid was made up to a total volume of 200 mL in a volumetric flask. A 25 mL portion of this acid was placed into a conical flask. During titration, 26.25 mL of 0.100 M sodium hydroxide was needed to neutralize the sample. Calculate the percentage purity of the carbonate sample.